
Assignment 7: Create a simple web service and write any distributed application to consume the web service.

Steps to Perform Calculator Web Service in Java using VS Code

Step 1: Open VS Code

Install:

Extension Pack for Java

Step 2: Create Project Folder

WebService

Open the folder in VS Code.

Step 3: Create Server File

Create file:

CalculatorServer.java

Paste the Calculator Server Java code and save the file.

Step 4: Create Client File

Create file:

CalculatorClient.java

Paste the Calculator Client Java code and save the file.

Step 5: Compile Programs

Open terminal in VS Code:

```
javac CalculatorServer.java  
javac CalculatorClient.java
```

Step 6: Run the Server

Open terminal 1:

```
java CalculatorServer
```

Output:

Server Started

Step 7: Run the Client

Open terminal 2:

```
java CalculatorClient
```

Expected Output

Server Side

Addition = 30

Client Side

Addition = 30

Viva Questions and Answers for Web Service Calculator Assignment

1. What is a Web Service?

A web service is a software component that allows communication between applications over a network.

2. What is the purpose of this assignment?

To create a calculator web service and access it through client-server communication.

3. Which language is used in your program?

Java

4. What is client-server architecture?

Client sends requests and server provides responses.

5. What is socket programming?

Socket programming enables communication between client and server applications.

6. Which classes are used in your program?

ServerSocket
Socket
DataInputStream
DataOutputStream

7. What is ServerSocket?

ServerSocket waits for client connection requests.

8. What is Socket?

Socket establishes communication between client and server.

9. Which port is used in your program?

5000

10. What is localhost?

Localhost refers to the current computer/server.

Example:

127.0.0.1

11. What is the output of your program?

Addition = 30

12. How do you run the server program?

java CalculatorServer

13. How do you run the client program?

java CalculatorClient

14. What is distributed application?

A distributed application runs on multiple systems connected through a network.

15. What are the calculator operations performed?

- Addition
- Subtraction
- Multiplication
- Division

16. Advantages of Web Services

- Platform independent
 - Easy communication
 - Reusable services
-

17. Disadvantages of Web Services

- Network dependency
 - Security issues
 - Slower than local applications
-

18. What is data communication?

Transfer of data between client and server systems.

19. What happens when client sends request?

Server receives request, processes it, and returns response.

20. Real-life applications of web services

- Banking systems
 - Cloud applications
 - Social media APIs
 - Payment gateways
-

21. Difference between website and web service

Website**Web Service**

Human interaction Application interaction

HTML pages Data/API response

22. What is HTTP?

HTTP stands for:

HyperText Transfer Protocol

Used for web communication.

23. Why are web services important?

They allow applications to communicate and share data easily.

24. What is input stream?

Input stream receives data from another system.

25. What is output stream?

Output stream sends data to another system.